State : mean x and distribution P

$$x_p^k = A x^{k-1} + B u^{k-1}$$

q^k process noise

prior to measurement

$$P_p^k = A P^{k-1} A^T + Q^k$$

covariance of q^k

sensor model

$$K^k = P_p^k H^T \left(H P_p^k H^T + R \right)^{-1}$$

covariance of r^k sensor noise

$$x^k = x_p^k + K^k \left(z^k - H x_p^k \right)$$

actual measurement

supposed to measure

$$P^k = P_p^k - K^k \cdot H \cdot P_p^k$$

$$x_p^k = A x^{k-1} + B u^{k-1}$$

$$P_p^k = A P^{k-1} A^T + Q$$

$$K^k = P_p^k H^T \left(H P_p^k H^T + R \right)^{-1}$$

$$x^k = x_p^k + K^k \left(z^k - H x_p^k \right)$$

$$P^k = P_p^k - K^k \cdot H \cdot P_p^k$$